

Iterative Decomposition

Decomposing a bundle into its constituents

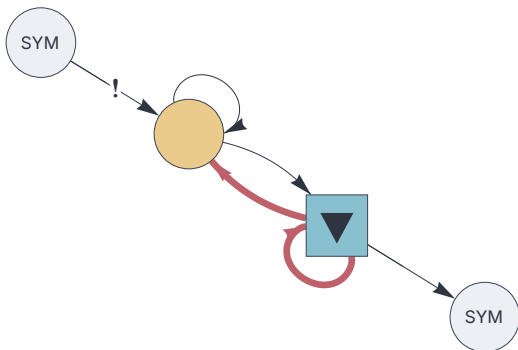
Decompose a bundle by repeatedly feeding it into an item memory. Each extracted item is removed from the recurrent network via inhibitory feedback.

```
<< "Circuits.m"
```

Circuit

```
dataflow =
{
  input[codec] [11] [],
  delay[] [21] [{11"!","21, 31"-}],
  auto["D"→0.8] [31] [{21, 31"-}],
  output[codec] [] [31],
  _ → {2000, 10}
};
```

```
c = Circuit[ dataflow]; c[Graph, ImageSize→ 280]
```



Train

```
items = CharacterRange["A", "Z"];  
c /@ items;  
c[Null]; (* flush delay *)
```

Analyze

```
c[{"A","B","C","D","E","F","G"}]; (* feed bundle of 7 items *)
```

```
Table[ c[Null], 16] (* trigger cycles with empty inputs *)
```

```
{Null, Null, B, C, A, D, F, E, G, Null, Null, Null, Null, Null, Null}
```